

# SOLIDWORKS API (.NET VBA C#) — Reference

## (226/337):

### IPrimaryMemberPathSegmentFeatureData-IPrint3DDialog

SOLIDWORKS API Documentation (converted from CHM)

# IPrimaryMemberPathSegmentFeatureData (Type)

## IPrimaryMemberPathSegmentFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IPrimaryMemberPathSegmentFeatureData Interface

Allows access to a structure system member created along path segments in a sketch.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface IPrimaryMemberPathSegmentFeatureData
```

Visual Basic (Usage)

```
Dim instance As IPrimaryMemberPathSegmentFeatureData
```

C#

```
public interface IPrimaryMemberPathSegmentFeatureData
```

C++/CLI

```
public interface class IPrimaryMemberPathSegmentFeatureData
```

### Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPathSegmentFeatureData](#).

### Example

'VBA

'This example creates primary structure system path segment members that are split along a reference plane.

```
'=====
'Preconditions:
'1. Ensure the specified part template exists.
'2. Open an Immediate window.
'3. Press F5.
'
'Postconditions:
' 1. Creates Sketch1 containing two sketch segments.
' 2. Creates Plane1.
' 3. Configures the start/end extensions and the member profile.
' 4. Selects two sketch segments and Plane1.
' 5. Configures structure system to split members using Plane1.
' 6. Creates a Structure System1 with four split members: Member1_1, Member1_2, Member2_1, and Member2_2.
' 7. Inspect the structure system in the graphics area.
' 8. Press F5 to edit the structure system.
' 9. Changes the start/end extensions for Member1_1 and Member1_2.
'10. Modifies the structure system.
'11. Inspect the changes in the graphics area.
'=====
```

```
Dim swApp As SldWorks.SldWorks
Dim modDoc As SldWorks.ModelDoc2
Dim swFeatMgr As SldWorks.FeatureManager
Dim swSelMgr As SldWorks.SelectionMgr
Dim modDocExt As SldWorks.ModelDocExtension
Dim structMemDef As SldWorks.StructureSystemMemberFeatureData
Dim profDef As SldWorks.StructureSystemMemberProfile
Dim PrimDef As SldWorks.PrimaryStructuralMemberFeatureData
Dim memPathSegDef As PrimaryMemberPathSegmentFeatureData
Dim segments(1) As Object
Dim stat As Boolean
Dim memData(0) As SldWorks.StructureSystemMemberFeatureData
Dim memDataArray As Variant
Dim structSys As SldWorks.Feature
```

```

Dim feat As SldWorks.Feature
Dim structSysModDef As SldWorks.StructureSystemFolder
Dim outProfiles As Variant
Dim MemberData As SldWorks.StructureSystemMemberFeatureData
Dim memberArray As Variant
Dim profileGrpFeat As SldWorks.Feature
Dim profileGrp As SldWorks.ProfileGroupFolder
Dim memTochange As SldWorks.StructureSystemMemberFeatureData
Dim i As Long
Dim j As Long
Option Explicit
Sub main()

    Set swApp = Application.SldWorks
    Dim swSheetWidth As Double
    swSheetWidth = 0
    Dim swSheetHeight As Double
    swSheetHeight = 0
    Set modDoc = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2022\templates\Part.PRTDOT", 0, swSheetWidth,
swSheetHeight)
    Set modDoc = swApp.ActiveDoc

    Set modDocExt = modDoc.Extension
    Set swSelMgr = modDoc.SelectionManager

    modDoc.SketchManager.InsertSketch True
    stat = modDocExt.SelectByID2("Front Plane", "PLANE", -5.50812963393624E-02, 2.28249965575248E-02, 1.35586835718293E-02, False, (
Nothing, 0)
    modDoc.ClearSelection2 True
    Dim skSegment As Object
    Set skSegment = modDoc.SketchManager.CreateLine(-0.036101, -0.017052, 0#, 0.03111, 0.018383, 0#)
    Set skSegment = modDoc.SketchManager.CreateLine(0.055067, -0.015222, 0#, -0.01281, -0.055483, 0#)

    modDoc.SketchManager.InsertSketch True
    modDoc.ClearSelection2 True

    stat = modDocExt.SelectByID2("Right Plane", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
    Dim myRefPlane As SldWorks.RefPlane
    Set myRefPlane = modDoc.FeatureManager.InsertRefPlane(8, 0.02, 0, 0, 0, 0)
    modDoc.ClearSelection2 True

    stat = modDocExt.SelectByID2("Line1@Sketch1", "EXTSKETCHSEGMENT", -1.23094293114227E-02, -4.50860223252535E-03, 0, True, 0,
Nothing, 0)
    stat = modDocExt.SelectByID2("Line2@Sketch1", "EXTSKETCHSEGMENT", 1.11728134432853, -3.76344266391984E-02, 0, True, 0,
Nothing, 0)
    stat = modDocExt.SelectByID2("Plane1", "PLANE", 1.99883401957978E-02, 2.02450835544425E-02, -2.43017856732353E-02, True, 0,
Nothing, 0)

    Set structMemDef = modDocExt.CreateStructureSystemMemberData(0)

    Debug.Print "Type of structure system member as defined by swStructureSystemMemberType_e: " &
structMemDef.StructureSystemMemberType

    structMemDef.StartEndExtendD1 = 0.1
    Debug.Print "Extend length in direction 1 (m): " & structMemDef.StartEndExtendD1
    structMemDef.StartEndExtendD2 = 0.2
    Debug.Print "Extend length in direction 2 (m): " & structMemDef.StartEndExtendD2

    Set profDef = structMemDef.MemberProfile

    profDef.ProfileStandard = "ansi inch"
    profDef.ProfileType = "c channel"
    profDef.ProfileSize = "3 x 5"

    Set PrimDef = structMemDef
    Debug.Print "Structure system primary member creation type as defined by swStructureSystemMemberCreationType_e: " &
PrimDef.PrimaryMemberType

    Set memPathSegDef = PrimDef
    memPathSegDef.MergeTangentMembers = True
    Debug.Print "Merge tangent members? " & memPathSegDef.MergeTangentMembers

    Set segments(0) = swSelMgr.GetSelectedObject6(1, 0)
    Set segments(1) = swSelMgr.GetSelectedObject6(2, 0)

    stat = memPathSegDef.SetPathSegments(segments)
    Debug.Print "Path segments set successfully: " & stat

    Dim planes(0) As Object
    Set planes(0) = swSelMgr.GetSelectedObject6(3, 0)
    structMemDef.IsSplit = True
    Dim structSplitMember As SldWorks.StructureSystemSplitMember
    Set structSplitMember = structMemDef.GetSplitMember

```

```

structSplitMember.MemberType = swStructureSplitMemberType_e.swStructureSplitMember_Reference
stat = structSplitMember.SetSplitReferences(planes)

Set memData(0) = structMemDef
memDataArray = memData

Set structSys = modDocExt.CreateStructureSystem(memDataArray, Nothing)

modDoc.ViewZoomtofit

Stop

'Edit the structure system
Set structSysModDef = structSys.GetSpecificFeature2
Debug.Print "Number of profile group folders: " & structSysModDef.GetProfileGroupFoldersCount

outProfiles = structSysModDef.GetProfileGroupFolders()

If (Not IsEmpty(outProfiles)) Then
    For l = 0 To UBound(outProfiles)
        Set profileGrpFeat = outProfiles(l)
        Set profileGrp = profileGrpFeat.GetSpecificFeature2
        If Not profileGrp Is Nothing Then
            Debug.Print "Number of members inside profile group folder: " & profileGrp.GetStructureSystemMemberCount
            memberArray = profileGrp.GetStructureSystemMembers()
            If (Not IsEmpty(memberArray)) Then
                For j = 0 To UBound(memberArray)
                    Set MemberData = memberArray(j) 'Get Member1 only
                    Exit For
                Next
            End If
        End If
    Next
End If

Set memToChange = MemberData

'Change the start/end extensions for Member1
Debug.Print "Change member extensions in directions 1 and 2 for one member only"
Set feat = memToChange.GetFeature()
memToChange.StartEndExtendD1 = 0.5
memToChange.StartEndExtendD2 = 0.2

'Save edits
stat = feat.ModifyDefinition(MemberData, modDoc, Nothing)

End Sub

```

### Example

[Create Path Segment Structure System Member \(VB.NET\)](#)  
[Create Path Segment Structure System Member \(C#\)](#)

### Remarks

See the [IStructureSystemFolder](#) Remarks.

For more information, see the **SOLIDWORKS Design user-interface help > Weldments and Structure System > Structure System > PrimaryMember PropertyManager** topic.

### Accessors

Cast [IPrimaryStructuralMemberFeatureData](#) or [IStructureSystemMemberFeatureData](#) to this interface.

### Access Diagram

[PrimaryMemberPathSegmentFeatureData](#)

### See Also

[IPrimaryMemberPathSegmentFeatureData Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : IPrimaryMemberPathSegmentFeatureData Interface

The following tables list the members exposed by [IPrimaryMemberPathSegmentFeatureData](#).

 **Public Properties**

| Name                                                                                                                  | Description                                        |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
|  <a href="#">MergeTangentMembers</a> | Gets and sets whether to merge tangential members. |

[Top](#)

 **Public Methods**

| Name                                                                                                                   | Description                                                                           |
|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  <a href="#">GetPathSegments</a>      | Gets the path segments, curves, and edges used to create the structure system member. |
|  <a href="#">GetPathSegmentsTypes</a> | Gets the types of path segments used to create the structure system member.           |
|  <a href="#">SetPathSegments</a>      | Gets the path segments, curves, and edges to create the structure system member.      |

[Top](#)

 **See Also**

[IPrimaryMemberPathSegmentFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

**IPrimaryMemberPathSegmentFeatureData Interface Methods**

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IPrimaryMemberPathSegmentFeatureData Interface

For a list of all members of this type, see [IPrimaryMemberPathSegmentFeatureData members](#).

 **Public Methods**

| Name                                                                                                                     | Description                                                                           |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  <a href="#">GetPathSegments</a>      | Gets the path segments, curves, and edges used to create the structure system member. |
|  <a href="#">GetPathSegmentsTypes</a> | Gets the types of path segments used to create the structure system member.           |
|  <a href="#">SetPathSegments</a>      | Gets the path segments, curves, and edges to create the structure system member.      |

[Top](#)

 **See Also**

[IPrimaryMemberPathSegmentFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

**IPrimaryMemberPathSegmentFeatureData Interface Properties**

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IPrimaryMemberPathSegmentFeatureData Interface

For a list of all members of this type, see [IPrimaryMemberPathSegmentFeatureData members](#).

 **Public Properties**

| Name                                                                                                                    | Description                                        |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
|  <a href="#">MergeTangentMembers</a> | Gets and sets whether to merge tangential members. |

[Top](#)

 **See Also**

[IPrimaryMemberPathSegmentFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

## GetPathSegments Method (IPrimaryMemberPathSegmentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberPathSegmentFeatureData Interface](#) : GetPathSegments Method (IPrimaryMemberPathSegmentFeatureData)

Gets the path segments, curves, and edges used to create the structure system member.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetPathSegments() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPathSegmentFeatureData  
Dim value As System.Object
```

```
value = instance.GetPathSegments()
```

C#

```
System.object GetPathSegments()
```

C++/CLI

```
System.Object^ GetPathSegments();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of [ISketchSegments](#), [IReferenceCurves](#), and/or [IEdges](#)

### Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPathSegmentFeatureData::GetPathSegments](#).

### See Also

[IPrimaryMemberPathSegmentFeatureData Interface](#)  
[IPrimaryMemberPathSegmentFeatureData Members](#)

### Availability

SOLIDWORKS 2022 FCS, Revision Number 30

## GetPathSegmentsTypes Method (IPrimaryMemberPathSegmentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberPathSegmentFeatureData Interface](#) : GetPathSegmentsTypes Method (IPrimaryMemberPathSegmentFeatureData)

Gets the types of path segments used to create the structure system member.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetPathSegmentsTypes() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPathSegmentFeatureData  
Dim value As System.Object
```

```
value = instance.GetPathSegmentsTypes()
```

C#

```
System.Object GetPathSegmentsTypes()
```

C++/CLI

```
System.Object^ GetPathSegmentsTypes();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of path segment types as defined by [swSelectType\\_e](#):

- swSelSKETCHES
- swSelEDGES
- swSelREFCURVES

### Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPathSegmentFeatureData::GetPathSegmentsTypes](#).

### See Also

[IPrimaryMemberPathSegmentFeatureData Interface](#)  
[IPrimaryMemberPathSegmentFeatureData Members](#)

### Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## MergeTangentMembers Property (IPrimaryMemberPathSegmentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberPathSegmentFeatureData Interface](#) : MergeTangentMembers Property (IPrimaryMemberPathSegmentFeatureData)

Gets and sets whether to merge tangential members.

### .NET Syntax

Visual Basic (Declaration)

```
Property MergeTangentMembers As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPathSegmentFeatureData  
Dim value As System.Boolean
```

```
instance.MergeTangentMembers = value
```

```
value = instance.MergeTangentMembers
```

C#

```
System.bool MergeTangentMembers {get; set;}
```

C++/CLI

```
property System.bool MergeTangentMembers {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True to merge tangential members, false to not

### Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPathSegmentFeatureData::MergeTangentMembers](#).

## Example

See the [IPrimaryMemberPathSegmentFeatureData](#) examples.

## Remarks

This property is valid only when creating the structure system path segment member.

## See Also

[IPrimaryMemberPathSegmentFeatureData Interface](#)  
[IPrimaryMemberPathSegmentFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## SetPathSegments Method (IPrimaryMemberPathSegmentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberPathSegmentFeatureData Interface](#) : SetPathSegments Method (IPrimaryMemberPathSegmentFeatureData)

Gets the path segments, curves, and edges to create the structure system member.

## .NET Syntax

Visual Basic (Declaration)

```
Function SetPathSegments( _  
    ByVal PathSegments As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPathSegmentFeatureData  
Dim PathSegments As System.Object  
Dim value As System.Boolean
```

```
value = instance.SetPathSegments(PathSegments)
```

C#

```
System.bool SetPathSegments(  
    System.object PathSegments  
)
```

C++/CLI

```
System.bool SetPathSegments(  
& System.Object^ PathSegments  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*PathSegments*  
Array of [ISketchSegments](#), [IReferenceCurves](#), and/or [IEdges](#)

### Return Value

True if path segments successfully set, false if not

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPathSegmentFeatureData::SetPathSegments](#).

### Example

See the [ICornerManagementFolder](#) examples.

See the [IPrimaryMemberPathSegmentFeatureData](#) examples.

### Remarks

For .NET applications, you must marshal the PathSegments array to IDispatch object arrays before calling this method. See [Dispatch Object Arrays as Input in .NET](#).

### See Also

[IPrimaryMemberPathSegmentFeatureData Interface](#)

[IPrimaryMemberPathSegmentFeatureData Members](#)

### Availability

SOLIDWORKS 2022 FCS, Revision Number 30

# IPrimaryMemberPointLengthFeatureData (Type)

## IPrimaryMemberPointLengthFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IPrimaryMemberPointLengthFeatureData Interface

Allows access to a structure system member originating at a point and extending to a specified end condition.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

Public Interface IPrimaryMemberPointLengthFeatureData

Visual Basic (Usage)

Dim instance As [IPrimaryMemberPointLengthFeatureData](#)

C#

public interface IPrimaryMemberPointLengthFeatureData

C++/CLI

public interface class IPrimaryMemberPointLengthFeatureData

### Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPointLengthFeatureData](#).

### Example

```
'VBA
'=====
'Preconditions:
'1. Ensure the specified part template exists.
'2. Open an Immediate window.
'3. Press F5.
'
'Postconditions:
'1. Creates Sketch1, Sketch2, and Sketch3 in a new part.
'2. Configures the start/end extensions and the member profile.
'3. Configures the length end condition for the structure members.
'4. Selects three sketch points.
'5. Specifies the length.
'6. Creates a Structure System1 with three structure system point length members.
'7. Inspect the structure system in the graphics area.
'8. Press F5 to edit the structure system.
'9. Gets one structure system member..
'10. Changes the length and member extensions in directions 1 and 2.
'11. Modifies the structure system.
'12. Inspect the changes in the graphics area.
'=====

Dim swApp As SldWorks.SldWorks
Dim modDoc As SldWorks.ModelDoc2
Dim swFeatMgr As SldWorks.FeatureManager
Dim swSelMgr As SldWorks.SelectionMgr
Dim modDocExt As SldWorks.ModelDocExtension
Dim structMemDef As SldWorks.StructureSystemMemberFeatureData
Dim profDef As SldWorks.StructureSystemMemberProfile
Dim PrimDef As SldWorks.PrimaryStructuralMemberFeatureData
Dim memPointLengthDef As SldWorks.PrimaryMemberPointLengthFeatureData
Dim startPoints(2) As Object
Dim stat As Boolean
Dim memData(0) As SldWorks.StructureSystemMemberFeatureData
Dim memDataArray As Variant
Dim systemFeature As SldWorks.Feature
Dim structSysModDef As SldWorks.StructureSystemFolder
```

```

Dim outProfiles As Variant
Dim MemberData As SldWorks.StructureSystemMemberFeatureData
Dim memberArray As Variant
Dim profileGrpFeat As SldWorks.Feature
Dim profileGrp As SldWorks.ProfileGroupFolder
Dim memTochange As SldWorks.StructureSystemMemberFeatureData
Dim pointLengthMember As SldWorks.PrimaryMemberPointLengthFeatureData
Dim memberFeat As SldWorks.Feature
Dim i As Long
Dim j As Long
Option Explicit
Sub main()

    Set swApp = Application.SldWorks
    Dim swSheetWidth As Double
    swSheetWidth = 0
    Dim swSheetHeight As Double
    swSheetHeight = 0
    Set modDoc = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2022\templates\Part.PRTDOT", 0, swSheetWidth,
swSheetHeight)
    Dim swPart As SldWorks.PartDoc
    Set swPart = modDoc
    Set modDoc = swApp.ActiveDoc

    Set modDocExt = modDoc.Extension
    Set swSelMgr = modDoc.SelectionManager

    modDoc.SketchManager.InsertSketch True
    stat = modDocExt.SelectByID2("Front Plane", "PLANE", -3.78286223842605E-02, 4.26130572878663E-02, -8.98903374939048E-03, False,
0, Nothing, 0)
    modDoc.ClearSelection2 True
    Dim skPoint As SldWorks.SketchPoint
    Set skPoint = modDoc.SketchManager.CreatePoint(-0.040799, 0.008094, 0#)
    modDoc.ClearSelection2 True
    modDoc.SketchManager.InsertSketch True
    stat = modDocExt.SelectByID2("Top Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    modDoc.SketchManager.InsertSketch True
    Set skPoint = modDoc.SketchManager.CreatePoint(-0.023183, 0.004864, 0#)
    modDoc.ClearSelection2 True
    modDoc.SketchManager.InsertSketch True
    stat = modDocExt.SelectByID2("Right Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    modDoc.SketchManager.InsertSketch True
    Set skPoint = modDoc.SketchManager.CreatePoint(-0.020877, 0.010539, 0#)
    modDoc.ClearSelection2 True
    modDoc.SketchManager.InsertSketch True

    modDoc.ClearSelection2 True
    modDoc.ShowNamedView2 "*Trimetric", 8
    modDoc.ViewZoomtofit2

    'Configure a structure system point length member
    Set structMemDef = modDocExt.CreateStructureSystemMemberData(2)

    Debug.Print "Type of structure system member as defined by swStructureSystemMemberType_e: " &
structMemDef.StructureSystemMemberType

    structMemDef.StartEndExtendD1 = 1#
    Debug.Print "Extend length in direction 1: " & structMemDef.StartEndExtendD1
    structMemDef.StartEndExtendD2 = 2#
    Debug.Print "Extend length in direction 2: " & structMemDef.StartEndExtendD2

    Set profDef = structMemDef.MemberProfile

    profDef.ProfileStandard = "ansi inch"
    profDef.ProfileType = "c channel"
    profDef.ProfileSize = "3 x 5"

    Set PrimDef = structMemDef
    Debug.Print "Structure system primary member creation type as defined by swStructureSystemMemberCreationType_e: " &
PrimDef.PrimaryMemberType; ""

    Set memPointLengthDef = PrimDef

    memPointLengthDef.EndCondition = 0

    'Set three points that define three structure system point length members
    stat = modDocExt.SelectByID2("Point1@Sketch3", "EXTSKETCHPOINT", -0.020877, 0.010539, 0, False, 0, Nothing, 0)
    stat = modDocExt.SelectByID2("Point1@Sketch1", "EXTSKETCHPOINT", -0.040799, 0.008094, 0, True, 0, Nothing, 0)
    stat = modDocExt.SelectByID2("Point1@Sketch2", "EXTSKETCHPOINT", -0.023183, 0.004864, 0, True, 0, Nothing, 0)

    Set startPoints(0) = swSelMgr.GetSelectedObject6(1, 0)
    Set startPoints(1) = swSelMgr.GetSelectedObject6(2, 0)

```

```

Set startPoints(2) = swSelMgr.GetSelectedObject6(3, 0)

stat = memPointLengthDef.SetPoints(startPoints)
Debug.Print "Start points set successfully: " & stat

'Set the length for each member
memPointLengthDef.Length = 5
memPointLengthDef.ReverseDirection = False

Set memData(0) = structMemDef

memDataArray = memData

Set systemFeature = modDocExt.CreateStructureSystem(memDataArray, Nothing)

modDoc.ViewZoomtofit2

Stop

Set structSysModDef = systemFeature.GetSpecificFeature2
Debug.Print "Number of structure system profile groups: " & structSysModDef.GetProfileGroupFoldersCount

outProfiles = structSysModDef.GetProfileGroupFolders()

If (Not IsEmpty(outProfiles)) Then
    For I = 0 To UBound(outProfiles)
        Set profileGrpFeat = outProfiles(I)
        Set profileGrp = profileGrpFeat.GetSpecificFeature2
        If Not profileGrp Is Nothing Then
            Debug.Print "Number of members inside profile group " & profileGrp.GetStructureSystemMemberCount
            memberArray = profileGrp.GetStructureSystemMembers()
            If (Not IsEmpty(memberArray)) Then
                For j = 0 To UBound(memberArray)
                    Set MemberData = memberArray(j) 'Get one structure system member to change
                    Exit For
                Next
            End If
        End If
    Next
End If

'Changing one member only
Set memToChange = MemberData

Debug.Print "Change member length and extensions in directions 1 and 2 for one member only"
memToChange.StartEndExtendD1 = 0.1
memToChange.StartEndExtendD2 = 0.1

Set pointLengthMember = MemberData
pointLengthMember.Length = 1

'Save edits
Set memberFeat = memToChange.GetFeature()
stat = memberFeat.ModifyDefinition(MemberData, modDoc, Nothing)

End Sub

```

#### Example

[Create Point Length Structure System Member \(VB.NET\)](#)  
[Create Point Length Structure System Member \(C#\)](#)

#### Remarks

See the [IStructureSystemFolder](#) Remarks.

For more information, see the **SOLIDWORKS Design** user-interface help > **Weldments and Structure System** > **Structure System** > **PrimaryMember PropertyManager** topic.

#### Accessors

Cast [IPrimaryStructuralMemberFeatureData](#) or [IStructureSystemMemberFeatureData](#) to this interface.

#### Access Diagram

[PrimaryMemberPointLengthFeatureData](#)

## See Also

[IPrimaryMemberPointLengthFeatureData Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## IPrimaryMemberPointLengthFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IPrimaryMemberPointLengthFeatureData Interface

The following tables list the members exposed by [IPrimaryMemberPointLengthFeatureData](#).

## Public Properties

| Name                                                                                                                     | Description                                                                           |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  <a href="#">DirectionReference</a>     | Gets and sets the reference indicating the direction of this member.                  |
|  <a href="#">EndCondition</a>           | Gets and sets the end condition for the member length.                                |
|  <a href="#">EndPlane</a>               | Gets and sets the end plane of this structure system member.                          |
|  <a href="#">EndPoint</a>               | Gets and sets the end point of this structure system member.                          |
|  <a href="#">Length</a>                 | Gets and sets the length of this structure system member.                             |
|  <a href="#">ReverseDirection</a>       | Gets and sets whether to reverse the length of this structure system member.          |
|  <a href="#">UnChainPointsAndLength</a> | Gets and sets whether to chain points to create this primary structure system member. |

[Top](#)

## Public Methods

| Name                                                                                                               | Description                                                                                         |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|  <a href="#">GetPoints</a>      | Gets the vertexes, sketch points, and reference points used to create this structure system member. |
|  <a href="#">GetPointsTypes</a> | Gets the types of points.                                                                           |
|  <a href="#">SetPoints</a>      | Sets the vertexes, sketch points, and reference points.                                             |

[Top](#)

## See Also

[IPrimaryMemberPointLengthFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## IPrimaryMemberPointLengthFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IPrimaryMemberPointLengthFeatureData Interface

For a list of all members of this type, see [IPrimaryMemberPointLengthFeatureData members](#).

## Public Methods

| Name                                                                                                               | Description                                                                                         |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|  <a href="#">GetPoints</a>      | Gets the vertexes, sketch points, and reference points used to create this structure system member. |
|  <a href="#">GetPointsTypes</a> | Gets the types of points.                                                                           |
|  <a href="#">SetPoints</a>      | Sets the vertexes, sketch points, and reference points.                                             |

[Top](#)

## See Also

[IPrimaryMemberPointLengthFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

# IPrimaryMemberPointLengthFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IPrimaryMemberPointLengthFeatureData Interface

For a list of all members of this type, see [IPrimaryMemberPointLengthFeatureData members](#).

## Public Properties

| Name                                                                                                                     | Description                                                                           |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  <a href="#">DirectionReference</a>     | Gets and sets the reference indicating the direction of this member.                  |
|  <a href="#">EndCondition</a>           | Gets and sets the end condition for the member length.                                |
|  <a href="#">EndPlane</a>               | Gets and sets the end plane of this structure system member.                          |
|  <a href="#">EndPoint</a>               | Gets and sets the end point of this structure system member.                          |
|  <a href="#">Length</a>                 | Gets and sets the length of this structure system member.                             |
|  <a href="#">ReverseDirection</a>       | Gets and sets whether to reverse the length of this structure system member.          |
|  <a href="#">UnChainPointsAndLength</a> | Gets and sets whether to chain points to create this primary structure system member. |

[Top](#)

## See Also

[IPrimaryMemberPointLengthFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## DirectionReference Property (IPrimaryMemberPointLengthFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberPointLengthFeatureData Interface](#) : DirectionReference Property (IPrimaryMemberPointLengthFeatureData)

Gets and sets the reference indicating the direction of this member.

## .NET Syntax

Visual Basic (Declaration)

Property DirectionReference As System.Object

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPointLengthFeatureData  
Dim value As System.Object
```

```
instance.DirectionReference = value
```

```
value = instance.DirectionReference
```

C#

```
System.object DirectionReference {get; set;}
```

C++/CLI

```
property System.Object^ DirectionReference {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

[IEdge](#) or [ISketchSegment](#)

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPointLengthFeatureData::DirectionReference](#).

## Remarks

This property is:

- required for 3D sketches.
- valid only when [IPrimaryMemberPointLengthFeatureData::EndCondition](#) is [swPrimaryMemberPointLengthEndCondition\\_e](#):
  - [swPrimaryMemberPointLengthEndCondition\\_UpToPlane](#)
  - [swPrimaryMemberPointLengthEndCondition\\_Length](#)

## See Also

[IPrimaryMemberPointLengthFeatureData Interface](#)  
[IPrimaryMemberPointLengthFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## EndCondition Property (IPrimaryMemberPointLengthFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberPointLengthFeatureData Interface](#) : EndCondition Property (IPrimaryMemberPointLengthFeatureData)

Gets and sets the end condition for the member length.

## .NET Syntax

Visual Basic (Declaration)

```
Property EndCondition As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPointLengthFeatureData  
Dim value As System.Integer
```

```
instance.EndCondition = value
```

```
value = instance.EndCondition
```

C#

```
System.int EndCondition {get; set;}
```

C++/CLI

```
property System.int EndCondition {  
    System.int get();  
    void set ( & System.int value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

End condition for member length as defined by [swPrimaryMemberPointLengthEndCondition\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPointLengthFeatureData::EndCondition](#).

## Example

See the [IPrimaryMemberPointLengthFeatureData](#) examples.

## See Also

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## EndPlane Property (IPrimaryMemberPointLengthFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberPointLengthFeatureData Interface](#) : EndPlane Property (IPrimaryMemberPointLengthFeatureData)

Gets and sets the end plane of this structure system member.

## .NET Syntax

### Visual Basic (Declaration)

Property EndPlane As System.Object

### Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPointLengthFeatureData  
Dim value As System.Object
```

```
instance.EndPlane = value
```

```
value = instance.EndPlane
```

### C#

```
System.object EndPlane {get; set;}
```

### C++/CLI

```
property System.Object^ EndPlane {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

[IRefPlane](#)

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPointLengthFeatureData::EndPlane](#).

## Remarks

This property is valid only if [IPrimaryMemberPointLengthFeatureData::EndCondition](#) is [swPrimaryMemberPointLengthEndCondition\\_e.swPrimaryMemberPointLengthEndCondition\\_UpToPlane](#).

## See Also

[IPrimaryMemberPointLengthFeatureData Interface](#)  
[IPrimaryMemberPointLengthFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## EndPoint Property (IPrimaryMemberPointLengthFeatureData)

[See Also](#)



Gets and sets the end point of this structure system member.

## .NET Syntax

### Visual Basic (Declaration)

```
Property EndPoint As System.Object
```

### Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPointLengthFeatureData
Dim value As System.Object

instance.EndPoint = value

value = instance.EndPoint
```

### C#

```
System.Object EndPoint {get; set;}
```

### C++/CLI

```
property System.Object^ EndPoint {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

[ISketchPoint](#)

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPointLengthFeatureData::EndPoint](#).

## Remarks

This property is valid only if [IPrimaryMemberPointLengthFeatureData::EndCondition](#) is [swPrimaryMemberPointLengthEndCondition\\_e.swPrimaryMemberPointLengthEndCondition\\_UpToPoint](#).

## See Also

[IPrimaryMemberPointLengthFeatureData Interface](#)  
[IPrimaryMemberPointLengthFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetPoints Method (IPrimaryMemberPointLengthFeatureData)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IPrimaryMemberPointLengthFeatureData Interface](#) : GetPoints Method (IPrimaryMemberPointLengthFeatureData)

Gets the vertexes, sketch points, and reference points used to create this structure system member.

## .NET Syntax

### Visual Basic (Declaration)

```
Function GetPoints() As System.Object
```

### Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPointLengthFeatureData
Dim value As System.Object

value = instance.GetPoints()
```

C#

```
System.object GetPoints()
```

C++/CLI

```
System.Object^ GetPoints();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of [IVertexes](#), [ISketchPoint](#)s, and/or [IRefPoint](#)s

### Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPointLengthFeatureData::GetPoints](#).

### See Also

[IPrimaryMemberPointLengthFeatureData Interface](#)  
[IPrimaryMemberPointLengthFeatureData Members](#)

### Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetPointsTypes Method (IPrimaryMemberPointLengthFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberPointLengthFeatureData Interface](#) : GetPointsTypes Method (IPrimaryMemberPointLengthFeatureData)

Gets the types of points.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetPointsTypes() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPointLengthFeatureData
Dim value As System.Object

value = instance.GetPointsTypes()
```

C#

```
System.object GetPointsTypes()
```

C++/CLI

```
System.Object^ GetPointsTypes();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of point types as defined by [swSelectType\\_e](#):

- swSelSKETCHPOINTS
- swSelEXTSKETCHPOINTS
- swSelVERTICES

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPointLengthFeatureData::GetPointsTypes](#).

## See Also

[IPrimaryMemberPointLengthFeatureData Interface](#)  
[IPrimaryMemberPointLengthFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments](#) on this topic.

## Length Property (IPrimaryMemberPointLengthFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberPointLengthFeatureData Interface](#) : Length Property (IPrimaryMemberPointLengthFeatureData)

Gets and sets the length of this structure system member.

## .NET Syntax

### Visual Basic (Declaration)

Property Length As System.Double

### Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPointLengthFeatureData  
Dim value As System.Double
```

```
instance.Length = value
```

```
value = instance.Length
```

### C#

```
System.double Length {get; set;}
```

### C++/CLI

```
property System.double Length {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Length

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPointLengthFeatureData::Length](#).

## Example

See the [IPrimaryMemberPointLengthFeatureData](#) examples.

## Remarks

This property is valid only if [IPrimaryMemberPointLengthFeatureData::EndCondition](#) is [swPrimaryMemberPointLengthEndCondition\\_e.swPrimaryMemberPointLengthEndCondition\\_Length](#).

## See Also

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## ReverseDirection Property (IPrimaryMemberPointLengthFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberPointLengthFeatureData Interface](#) : ReverseDirection Property (IPrimaryMemberPointLengthFeatureData)

Gets and sets whether to reverse the length of this structure system member.

## .NET Syntax

### Visual Basic (Declaration)

Property ReverseDirection As System.Boolean

### Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPointLengthFeatureData  
Dim value As System.Boolean
```

```
instance.ReverseDirection = value
```

```
value = instance.ReverseDirection
```

### C#

```
System.bool ReverseDirection {get; set;}
```

### C++/CLI

```
property System.bool ReverseDirection {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True to reverse the member direction, false to not

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPointLengthFeatureData.ReverseDirection](#).

## Example

See the [IPrimaryMemberPointLengthFeatureData](#) examples.

## Remarks

This property is valid only if [IPrimaryMemberPointLengthFeatureData::EndCondition](#) is [swPrimaryMemberPointLengthEndCondition\\_e.swPrimaryMemberPointLengthEndCondition\\_Length](#).

## See Also

[IPrimaryMemberPointLengthFeatureData Interface](#)  
[IPrimaryMemberPointLengthFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

## SetPoints Method (IPrimaryMemberPointLengthFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberPointLengthFeatureData Interface](#) : SetPoints Method (IPrimaryMemberPointLengthFeatureData)

Sets the vertexes, sketch points, and reference points.

### .NET Syntax

#### Visual Basic (Declaration)

```
Function SetPoints( _  
    ByVal Points As System.Object _  
) As System.Boolean
```

#### Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPointLengthFeatureData  
Dim Points As System.Object  
Dim value As System.Boolean
```

```
value = instance.SetPoints(Points)
```

#### C#

```
System.bool SetPoints(  
    System.object Points  
)
```

#### C++/CLI

```
System.bool SetPoints(  
& System.Object^ Points  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

##### *Points*

Array of [IVertexes](#), [ISketchPoints](#), and/or [IRefPoints](#)

#### Return Value

True if points successfully set, false if not

### Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPointLengthFeatureData.SetPoints](#).

### Example

See the [IPrimaryMemberPointLengthFeatureData](#) examples.

### Remarks

Use [IPrimaryMemberPointLengthFeatureData::UnChainPointsAndLength](#) to specify whether to chain link members along a chain of points.

### See Also

[IPrimaryMemberPointLengthFeatureData Interface](#)  
[IPrimaryMemberPointLengthFeatureData Members](#)

### Availability

SOLIDWORKS 2022 FCS, Revision Number 30

## UnChainPointsAndLength Property (IPrimaryMemberPointLengthFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberPointLengthFeatureData Interface](#) : UnChainPointsAndLength Property (IPrimaryMemberPointLengthFeatureData)

Gets and sets whether to chain points to create this primary structure system member.

### .NET Syntax

Visual Basic (Declaration)

Property UnChainPointsAndLength As System.Boolean

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberPointLengthFeatureData  
Dim value As System.Boolean
```

```
instance.UnChainPointsAndLength = value
```

```
value = instance.UnChainPointsAndLength
```

C#

```
System.bool UnChainPointsAndLength {get; set;}
```

C++/CLI

```
property System.bool UnChainPointsAndLength {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True to pair each point passed in [IPrimaryMemberPointLengthFeatureData::SetPoints](#) to the previous point in the list, false to pair consecutive points only

### Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberPointLengthFeatureData::UnChainPointsAndLength](#).

### Remarks

This property is valid only if [IPrimaryMemberPointLengthFeatureData::EndCondition](#) is [swPrimaryMemberPointLengthEndCondition\\_e.swPrimaryMemberPointLengthEndCondition\\_Point](#).

If this property is set to true, each point passed to [IPrimaryMemberPointLengthFeatureData::SetPoints](#) pairs with the previous point in the list to create a new member.

For the list of points A,B,C,D,E,F:

If this property is set to true, then 5 members are created:

A -- B

B -- C

C -- D

D -- E

E -- F

If this property is set to false, then only 3 members are created:

A -- B

C -- D

E -- F

### See Also

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

# IPrimaryMemberRefPlaneFeatureData (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## IPrimaryMemberRefPlaneFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IPrimaryMemberRefPlaneFeatureData Interface

Allows access to a structure system member created along the intersection of two or more planes.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface IPrimaryMemberRefPlaneFeatureData
```

Visual Basic (Usage)

```
Dim instance As IPrimaryMemberRefPlaneFeatureData
```

C#

```
public interface IPrimaryMemberRefPlaneFeatureData
```

C++/CLI

```
public interface class IPrimaryMemberRefPlaneFeatureData
```

### Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberRefPlaneFeatureData](#).

### Example

'VBA

```
'=====
'Preconditions:
'1. Ensure the specified part template exists.
'2. Open an Immediate window.
'3. Press F5.
'
'Postconditions:
'1. Creates five reference planes:
'   - Plane1 parallel to Right Plane
'   - Plane2 and Plane3 parallel to and flanking Top Plane
'   - Plane4 and Plane5 parallel to and flanking Front Plane
'2. Configures the member profile.
'3. Selects the start/end reference planes that define the lengths of the members: Plane1 and Right Plane.
'4. Selects the axis reference planes that define the direction of the members: Plane2 and Plane3.
'5. Selects the location reference planes that define the positions of the members: Plane4 and Plane5.
'6. Creates a Structure System1 with nine structure system reference plane members.
'7. Inspect the structure system in the graphics area.
'8. Press F5 to edit the structure system.
'9. Gets Member1.
'10. Changes its extensions in directions 1 and 2.
'11. Modifies the structure system.
'12. Inspect the changes in the graphics area.
'=====
```

```
Dim swApp As SldWorks.SldWorks
Dim modDoc As SldWorks.ModelDoc2
Dim swFeatMgr As SldWorks.FeatureManager
Dim swSelMgr As SldWorks.SelectionMgr
Dim modDocExt As SldWorks.ModelDocExtension
Dim structMemDef As SldWorks.StructureSystemMemberFeatureData
Dim profDef As SldWorks.StructureSystemMemberProfile
Dim PrimDef As SldWorks.PrimaryStructuralMemberFeatureData
Dim memRefPlaneDef As PrimaryMemberRefPlaneFeatureData
Dim startEndPlanes(1) As Object
Dim horzPlanes(2) As Object
Dim vertPlanes(2) As Object
```

```

Dim stat As Boolean
Dim memData(0) As SldWorks.StructureSystemMemberFeatureData
Dim memDataArray As Variant
Dim structSysFeat As SldWorks.Feature
Dim feat As SldWorks.Feature
Dim structSysModDef As SldWorks.StructureSystemFolder
Dim outProfiles As Variant
Dim MemberData As SldWorks.StructureSystemMemberFeatureData
Dim memberArray As Variant
Dim profileGrpFeat As SldWorks.Feature
Dim profileGrp As SldWorks.ProfileGroupFolder
Dim memToChange As SldWorks.StructureSystemMemberFeatureData
Dim i As Long
Dim j As Long
Option Explicit
Sub main()

    Set swApp = Application.SldWorks

    Dim swSheetWidth As Double
    swSheetWidth = 0
    Dim swSheetHeight As Double
    swSheetHeight = 0
    Set modDoc = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2022\templates\Part.PRTDOT", 0, swSheetWidth,
swSheetHeight)

    Set modDoc = swApp.ActiveDoc

    Set swSelMgr = modDoc.SelectionManager
    Set modDocExt = modDoc.Extension

    stat = modDocExt.SelectByID2("Right Plane", "PLANE", 0, 0, 0, True, 0, Nothing, 0)

    Dim myRefPlane As SldWorks.RefPlane
    Set myRefPlane = modDoc.FeatureManager.InsertRefPlane(8, 0.381, 0, 0, 0, 0)
    modDoc.ClearSelection2 True
    stat = modDocExt.SelectByID2("Top Plane", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
    Set myRefPlane = modDoc.FeatureManager.InsertRefPlane(8, 0.1778, 0, 0, 0, 0)
    modDoc.ClearSelection2 True
    stat = modDocExt.SelectByID2("Plane1", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    modDoc.ClearSelection2 True
    stat = modDocExt.SelectByID2("Top Plane", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
    Set myRefPlane = modDoc.FeatureManager.InsertRefPlane(264, 0.1778, 0, 0, 0, 0)
    modDoc.ClearSelection2 True
    stat = modDocExt.SelectByID2("Plane2", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    modDoc.ClearSelection2 True
    stat = modDocExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
    Set myRefPlane = modDoc.FeatureManager.InsertRefPlane(8, 0.1778, 0, 0, 0, 0)
    modDoc.ClearSelection2 True
    stat = modDocExt.SelectByID2("Plane3", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    modDoc.ClearSelection2 True
    stat = modDocExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
    Set myRefPlane = modDoc.FeatureManager.InsertRefPlane(264, 0.1778, 0, 0, 0, 0)
    modDoc.ClearSelection2 True
    stat = modDocExt.SelectByID2("Plane4", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    modDoc.ClearSelection2 True

    stat = modDocExt.SelectByID2("Right Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    stat = modDocExt.SelectByID2("Plane1", "PLANE", 7.29275826275853E-02, 0.103693179609365, 0.188906126602433, True, 0, Nothing, 0)
    stat = modDocExt.SelectByID2("Plane2", "PLANE", -0.035035924945646, 8.82769827595098E-02, 0.154770056203461, True, 0, Nothing, 0)
    stat = modDocExt.SelectByID2("Top Plane", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
    stat = modDocExt.SelectByID2("Plane3", "PLANE", -3.59335022268397E-02, 4.86897858681345E-02, 0.154489603864079, True, 0, Nothing,
0)
    stat = modDocExt.SelectByID2("Plane4", "PLANE", -3.53613843872154E-02, 0.122848840931965, 0.120060929442332, True, 0, Nothing, 0)
    stat = modDocExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
    stat = modDocExt.SelectByID2("Plane5", "PLANE", -3.13617150023902E-02, 0.123208535607716, 7.98165581882131E-02, True, 0, Nothing,
0)

    'Create structure system reference plane member
    Set structMemDef = modDocExt.CreateStructureSystemMemberData(1)
    Debug.Print "Type of structure system member as defined by swStructureSystemMemberType_e: " &
structMemDef.StructureSystemMemberType

    Debug.Print "Extend length in direction 1 (m): " & structMemDef.StartEndExtendD1
    Debug.Print "Extend length in direction 2 (m): " & structMemDef.StartEndExtendD2

    Set profDef = structMemDef.MemberProfile

    profDef.ProfileStandard = "ansi inch"
    profDef.ProfileType = "c channel"
    profDef.ProfileSize = "3 x 5"

```

```

Set PrimDef = structMemDef
Debug.Print "Structure system primary member creation type as defined by swStructureSystemMemberCreationType_e: " &
PrimDef.PrimaryMemberType; ""

Set memRefPlaneDef = PrimDef

'Define the member lengths
Set startEndPlanes(0) = swSelMgr.GetSelectedObject6(1, 0)
Set startEndPlanes(1) = swSelMgr.GetSelectedObject6(2, 0)

stat = memRefPlaneDef.SetStartAndEndReferences(startEndPlanes)
Debug.Print "Reference planes set successfully: " & stat

Set horzPlanes(0) = swSelMgr.GetSelectedObject6(3, 0)
Set horzPlanes(1) = swSelMgr.GetSelectedObject6(4, 0)
Set horzPlanes(2) = swSelMgr.GetSelectedObject6(5, 0)

stat = memRefPlaneDef.SetReferenceAxes(horzPlanes)
Debug.Print "Reference axis planes set successfully: " & stat

Set vertPlanes(0) = swSelMgr.GetSelectedObject6(6, 0)
Set vertPlanes(1) = swSelMgr.GetSelectedObject6(7, 0)
Set vertPlanes(2) = swSelMgr.GetSelectedObject6(8, 0)

stat = memRefPlaneDef.SetReferenceLocations(vertPlanes)
Debug.Print "Reference location planes set successfully: " & stat

Set memData(0) = structMemDef

memDataArray = memData

Set structSysFeat = modDocExt.CreateStructureSystem(memDataArray, Nothing)

modDoc.ShowNamedView2 "**Trimetric", 8
modDoc.ViewZoomtofit2

Stop

Set structSysModDef = structSysFeat.GetSpecificFeature2
Debug.Print "Number of profile group folders: " & structSysModDef.GetProfileGroupFoldersCount

outProfiles = structSysModDef.GetProfileGroupFolders()

If (Not IsEmpty(outProfiles)) Then
    For I = 0 To UBound(outProfiles)
        Set profileGrpFeat = outProfiles(I)
        Set profileGrp = profileGrpFeat.GetSpecificFeature2
        If Not profileGrp Is Nothing Then
            Debug.Print "Number of members in profile group folder: " & profileGrp.GetStructureSystemMemberCount
            memberArray = profileGrp.GetStructureSystemMembers()
            If (Not IsEmpty(memberArray)) Then
                For j = 0 To UBound(memberArray)
                    Set MemberData = memberArray(j) 'Get first member only
                    Exit For
                Next
            End If
        End If
    Next
End If

Set memTochange = MemberData

Set feat = memTochange.GetFeature()

'Change the extensions in directions 1 and 2 for Member1
memTochange.StartEndExtendD1 = 0.5
memTochange.StartEndExtendD2 = 0.2

modDoc.ClearSelection2 True

Debug.Print "Member extensions set successfully: " & stat

'Save edits
stat = feat.ModifyDefinition(MemberData, modDoc, Nothing)

End Sub

```

#### Example

[Create Reference Plane Structure System Member \(VB.NET\)](#)  
[Create Reference Plane Structure System Member \(C#\)](#)

## Remarks

The number of members created = ([IPrimaryMemberRefPlaneFeatureData::GetReferenceAxes](#) \* [IPrimaryMemberRefPlaneFeatureData::GetReferenceLocations](#))

See the [IStructureSystemFolder](#) Remarks.

For more information, see the **SOLIDWORKS Design user-interface help > Weldments and Structure System > Structure System > PrimaryMember PropertyManager** topic.

## Accessors

Cast [IPrimaryStructuralMemberFeatureData](#) or [IStructureSystemMemberFeatureData](#) to this interface.

## Access Diagram

[PrimaryMemberRefPlaneFeatureData](#)

## See Also

[IPrimaryMemberRefPlaneFeatureData Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## IPrimaryMemberRefPlaneFeatureData Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IPrimaryMemberRefPlaneFeatureData Interface

The following tables list the members exposed by [IPrimaryMemberRefPlaneFeatureData](#).

## Public Methods

| Name                                                                                                                              | Description                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|  <a href="#">GetReferenceAxes</a>              | Gets the direction reference(s) used to create this structure system member.                                  |
|  <a href="#">GetReferenceAxesTypes</a>         | Gets the types of reference axes.                                                                             |
|  <a href="#">GetReferenceLocations</a>         | Gets the position reference(s) used to create this structure system member.                                   |
|  <a href="#">GetReferenceLocationsTypes</a>    | Gets the types of location references for this structure system member.                                       |
|  <a href="#">GetStartAndEndReferences</a>      | Gets the start and end reference entities that are used to define the length of this structure system member. |
|  <a href="#">GetStartAndEndReferencesTypes</a> | Gets the types of the start and end reference entities of this structure system member.                       |
|  <a href="#">SetReferenceAxes</a>              | Sets the direction reference(s) for this structure system member.                                             |
|  <a href="#">SetReferenceLocations</a>         | Gets the location reference(s) used to create this structure system member.                                   |
|  <a href="#">SetStartAndEndReferences</a>      | Sets the start and end reference entities that are used to define the length of this structure system member. |

[Top](#)

## See Also

[IPrimaryMemberRefPlaneFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## IPrimaryMemberRefPlaneFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IPrimaryMemberRefPlaneFeatureData Interface

For a list of all members of this type, see [IPrimaryMemberRefPlaneFeatureData members](#).

## Public Methods

| Name                                                                                                                 | Description                                                                  |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
|  <a href="#">GetReferenceAxes</a> | Gets the direction reference(s) used to create this structure system member. |

|                                                                                   |                                               |                                                                                                               |
|-----------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|  | <a href="#">GetReferenceAxesTypes</a>         | Gets the types of reference axes.                                                                             |
|  | <a href="#">GetReferenceLocations</a>         | Gets the position reference(s) used to create this structure system member.                                   |
|  | <a href="#">GetReferenceLocationsTypes</a>    | Gets the types of location references for this structure system member.                                       |
|  | <a href="#">GetStartAndEndReferences</a>      | Gets the start and end reference entities that are used to define the length of this structure system member. |
|  | <a href="#">GetStartAndEndReferencesTypes</a> | Gets the types of the start and end reference entities of this structure system member.                       |
|  | <a href="#">SetReferenceAxes</a>              | Sets the direction reference(s) for this structure system member.                                             |
|  | <a href="#">SetReferenceLocations</a>         | Gets the location reference(s) used to create this structure system member.                                   |
|  | <a href="#">SetStartAndEndReferences</a>      | Sets the start and end reference entities that are used to define the length of this structure system member. |

[Top](#)

## See Also

[IPrimaryMemberRefPlaneFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetReferenceAxes Method (IPrimaryMemberRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberRefPlaneFeatureData Interface](#) : GetReferenceAxes Method (IPrimaryMemberRefPlaneFeatureData)

Gets the direction reference(s) used to create this structure system member.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetReferenceAxes() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberRefPlaneFeatureData
Dim value As System.Object

value = instance.GetReferenceAxes()
```

C#

```
System.object GetReferenceAxes()
```

C++/CLI

```
System.Object^ GetReferenceAxes();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Array of parallel [IRefPlanes](#) and/or [IFace2s](#)

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberRefPlaneFeatureData::GetReferenceAxes](#).

## Remarks

The parallel entities returned by this method must be perpendicular to and intersect the start and end references.

The number of members created = (IPrimaryMemberRefPlaneFeatureData::GetReferenceAxes

\* [IPrimaryMemberRefPlaneFeatureData::GetReferenceLocations](#))

## See Also

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetReferenceAxesTypes Method (IPrimaryMemberRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberRefPlaneFeatureData Interface](#) : GetReferenceAxesTypes Method (IPrimaryMemberRefPlaneFeatureData)

Gets the types of reference axes.

## .NET Syntax

### Visual Basic (Declaration)

```
Function GetReferenceAxesTypes() As System.Object
```

### Visual Basic (Usage)



```
Dim instance As IPrimaryMemberRefPlaneFeatureData  
Dim value As System.Object
```

```
value = instance.GetReferenceAxesTypes()
```

### C#

```
System.Object GetReferenceAxesTypes()
```

### C++/CLI

```
System.Object^ GetReferenceAxesTypes();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Array of start and end reference types as defined by [swSelectType\\_e](#):

- swSelDATUMPLANES
- swSelFACES

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberRefPlaneFeatureData::GetReferenceAxesTypes](#).

## See Also

[IPrimaryMemberRefPlaneFeatureData Interface](#)  
[IPrimaryMemberRefPlaneFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetReferenceLocations Method (IPrimaryMemberRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberRefPlaneFeatureData Interface](#) : GetReferenceLocations Method (IPrimaryMemberRefPlaneFeatureData)

Gets the position reference(s) used to create this structure system member.

## .NET Syntax

### Visual Basic (Declaration)

## Function GetReferenceLocations() As System.Object

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberRefPlaneFeatureData  
Dim value As System.Object
```

```
value = instance.GetReferenceLocations()
```

C#

```
System.object GetReferenceLocations()
```

C++/CLI

```
System.Object^ GetReferenceLocations();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Array of parallel [IRefPlanes](#) and/or [IFace2s](#)

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberRefPlaneFeatureData::GetReferenceLocations](#).

## Remarks

The parallel entities returned by this method can be parallel or non-parallel to the start and end reference entities.

The number of members created = ([IPrimaryMemberRefPlaneFeatureData::GetReferenceAxes](#)  
\* [IPrimaryMemberRefPlaneFeatureData::GetReferenceLocations](#))

## See Also

[IPrimaryMemberRefPlaneFeatureData Interface](#)  
[IPrimaryMemberRefPlaneFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetReferenceLocationsTypes Method (IPrimaryMemberRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberRefPlaneFeatureData Interface](#) : GetReferenceLocationsTypes Method (IPrimaryMemberRefPlaneFeatureData)

Gets the types of location references for this structure system member.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetReferenceLocationsTypes() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberRefPlaneFeatureData  
Dim value As System.Object
```

```
value = instance.GetReferenceLocationsTypes()
```

C#

```
System.Object GetReferenceLocationsTypes()
```

C++/CLI

```
System.Object^ GetReferenceLocationsTypes();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of start and end reference types as defined by [swSelectType\\_e](#):

- swSelDATUMPLANES
- swSelFACES

### Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberRefPlaneFeatureData::GetReferenceLocationsTypes](#).

### See Also

[IPrimaryMemberRefPlaneFeatureData Interface](#)  
[IPrimaryMemberRefPlaneFeatureData Members](#)

### Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetStartAndEndReferences Method (IPrimaryMemberRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberRefPlaneFeatureData Interface](#) : GetStartAndEndReferences Method (IPrimaryMemberRefPlaneFeatureData)

Gets the start and end reference entities that are used to define the length of this structure system member.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetStartAndEndReferences() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberRefPlaneFeatureData  
Dim value As System.Object
```

```
value = instance.GetStartAndEndReferences()
```

C#

```
System.Object GetStartAndEndReferences()
```

C++/CLI

```
System.Object^ GetStartAndEndReferences();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of two parallel entities ([IRefPlanes](#) and/or [IFace2s](#))

### Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberRefPlaneFeatureData::GetStartAndEndReferences](#).

### Remarks

The parallel entities returned by this method must be perpendicular and intersect the reference axes.

## See Also

[IPrimaryMemberRefPlaneFeatureData Interface](#)  
[IPrimaryMemberRefPlaneFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetStartAndEndReferencesTypes Method (IPrimaryMemberRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberRefPlaneFeatureData Interface](#) : GetStartAndEndReferencesTypes Method (IPrimaryMemberRefPlaneFeatureData)

Gets the types of the start and end reference entities of this structure system member.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetStartAndEndReferencesTypes() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberRefPlaneFeatureData  
Dim value As System.Object  
  
value = instance.GetStartAndEndReferencesTypes()
```

C#

```
System.object GetStartAndEndReferencesTypes()
```

C++/CLI

```
System.Object^ GetStartAndEndReferencesTypes();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Array of start and end reference types as defined by [swSelectType\\_e](#):

- swSelDATUMPLANES
- swSelFACES

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberRefPlaneFeatureData::GetStartAndEndReferencesTypes](#).

## See Also

[IPrimaryMemberRefPlaneFeatureData Interface](#)  
[IPrimaryMemberRefPlaneFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## SetReferenceAxes Method (IPrimaryMemberRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberRefPlaneFeatureData Interface](#) : SetReferenceAxes Method (IPrimaryMemberRefPlaneFeatureData)

Sets the direction reference(s) for this structure system member.

## Visual Basic Syntax

### Visual Basic (Declaration)

```
Function SetReferenceAxes( _  
    ByVal RetEntities As System.Object _  
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IPrimaryMemberRefPlaneFeatureData  
Dim RetEntities As System.Object  
Dim value As System.Boolean  
  
value = instance.SetReferenceAxes(RetEntities)
```

### C#

```
System.bool SetReferenceAxes(  
    System.object RetEntities  
)
```

### C++/CLI

```
System.bool SetReferenceAxes(  
& System.Object^ RetEntities  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*RetEntities*  
Array of parallel [IRefPlanes](#) and/or [IFace2s](#)

#### Return Value

True if reference axes successfully set, false if not

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberRefPlaneFeatureData::SetReferenceAxes](#).

## Example

See the [IPrimaryMemberRefPlaneFeatureData](#) examples.

## Remarks

The number of members created = ([IPrimaryMemberRefPlaneFeatureData::GetReferenceAxes](#)  
\* [IPrimaryMemberRefPlaneFeatureData::GetReferenceLocations](#))

During:

- creation, you can set any number of entities that are intersecting with the start and end reference ([IPrimaryMemberRefPlaneFeatureData::GetStartAndEndReferences](#)).
- editing, you can set only one entity that intersects with the start and end reference.

## See Also

[IPrimaryMemberRefPlaneFeatureData Interface](#)  
[IPrimaryMemberRefPlaneFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30



Gets the location reference(s) used to create this structure system member.

## .NET Syntax

### Visual Basic (Declaration)

```
Function SetReferenceLocations( _
    ByVal RetEntities As System.Object _
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IPrimaryMemberRefPlaneFeatureData
Dim RetEntities As System.Object
Dim value As System.Boolean

value = instance.SetReferenceLocations(RetEntities)
```

### C#

```
System.bool SetReferenceLocations(
    System.object RetEntities
)
```

### C++/CLI

```
System.bool SetReferenceLocations(
    & System.Object^ RetEntities
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*RetEntities*

Array of parallel [IRefPlanes](#) and/or [IFace2s](#)

#### Return Value

True if location references successfully set, false if not

## Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberRefPlaneFeatureData::SetReferenceLocations](#).

## Example

See the [IPrimaryMemberRefPlaneFeatureData](#) examples.

## Remarks

The number of members created = ([IPrimaryMemberRefPlaneFeatureData::GetReferenceAxes](#)  
\* [IPrimaryMemberRefPlaneFeatureData::GetReferenceLocations](#))

During:

- creation, you can set any number of entities that are intersecting with the start and end references and at least one reference axis entity.
- editing, you can set only one entity that is intersecting with the start and end references and one reference axis entity.

## See Also

[IPrimaryMemberRefPlaneFeatureData Interface](#)  
[IPrimaryMemberRefPlaneFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

## SetStartAndEndReferences Method (IPrimaryMemberRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryMemberRefPlaneFeatureData Interface](#) : SetStartAndEndReferences Method (IPrimaryMemberRefPlaneFeatureData)

Sets the start and end reference entities that are used to define the length of this structure system member.

### .NET Syntax

Visual Basic (Declaration)

```
Function SetStartAndEndReferences( _  
    ByVal RetEntities As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPrimaryMemberRefPlaneFeatureData  
Dim RetEntities As System.Object  
Dim value As System.Boolean
```

```
value = instance.SetStartAndEndReferences(RetEntities)
```

C#

```
System.bool SetStartAndEndReferences(  
    System.object RetEntities  
)
```

C++/CLI

```
System.bool SetStartAndEndReferences(  
& System.Object^ RetEntities  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*RetEntities*

Array of parallel [IRefPlanes](#) and/or [IFace2s](#)

#### Return Value

True if start and end references successfully set, false if not

### Visual Basic for Applications (VBA) Syntax

See [PrimaryMemberRefPlaneFeatureData::SetStartAndEndReferences](#).

### Example

See the [IPrimaryMemberRefPlaneFeatureData](#) examples.

### Remarks

The array set by this method should have only two entities: start and end references.

### See Also

[IPrimaryMemberRefPlaneFeatureData Interface](#)  
[IPrimaryMemberRefPlaneFeatureData Members](#)

### Availability

SOLIDWORKS 2022 FCS, Revision Number 30

# IPrimaryStructuralMemberFeatureData (Type)

## IPrimaryStructuralMemberFeatureData Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : IPrimaryStructuralMemberFeatureData Interface

Allows access to a primary structure system member.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### Visual Basic Syntax

Visual Basic (Declaration)

```
Public Interface IPrimaryStructuralMemberFeatureData
```

Visual Basic (Usage)



```
Dim instance As IPrimaryStructuralMemberFeatureData
```

C#

```
public interface IPrimaryStructuralMemberFeatureData
```

C++/CLI

```
public interface class IPrimaryStructuralMemberFeatureData
```

### Visual Basic for Applications (VBA) Syntax

See [PrimaryStructuralMemberFeatureData](#).

### Example

See the [IPrimaryMemberFacePlaneIntersectionFeatureData](#) examples.

See the [IPrimaryMemberPointLengthFeatureData](#) examples.

See the [IPrimaryMemberPathSegmentFeatureData](#) examples.

See the [IPrimaryMemberRefPlaneFeatureData](#) examples.

### Remarks

This interface is a superclass of the following interfaces, i.e., you can cast this interface to any of the following:

- [IPrimaryMemberFacePlaneIntersectionFeatureData](#)
- [IPrimaryMemberPathSegmentFeatureData](#)
- [IPrimaryMemberPointLengthFeatureData](#)
- [IPrimaryMemberRefPlaneFeatureData](#)

See the [IStructureSystemFolder](#) Remarks.

### Accessors

Cast [IStructureSystemMemberFeatureData](#) to this interface.

### Access Diagram

[PrimaryStructuralMemberFeatureData](#)

### See Also

[IPrimaryStructuralMemberFeatureData Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)

[See Also](#) [Properties](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IPrimaryStructuralMemberFeatureData Interface

The following tables list the members exposed by [IPrimaryStructuralMemberFeatureData](#).

 **Public Properties**

| Name                                                                                                                | Description                                            |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
|  <a href="#">PrimaryMemberType</a> | Gets the type of this primary structure system member. |

[Top](#)

 **See Also**

[IPrimaryStructuralMemberFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

**IPrimaryStructuralMemberFeatureData Interface Properties**

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IPrimaryStructuralMemberFeatureData Interface

For a list of all members of this type, see [IPrimaryStructuralMemberFeatureData members](#).

 **Public Properties**

| Name                                                                                                                | Description                                            |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
|  <a href="#">PrimaryMemberType</a> | Gets the type of this primary structure system member. |

[Top](#)

 **See Also**

[IPrimaryStructuralMemberFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

**PrimaryMemberType Property (IPrimaryStructuralMemberFeatureData)**

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPrimaryStructuralMemberFeatureData Interface](#) : PrimaryMemberType Property (IPrimaryStructuralMemberFeatureData)

Gets the type of this primary structure system member.

 **.NET Syntax**

Visual Basic (Declaration)

```
ReadOnly Property PrimaryMemberType As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IPrimaryStructuralMemberFeatureData  
Dim value As System.Integer  
  
value = instance.PrimaryMemberType
```

C#

```
System.int PrimaryMemberType {get;}
```

C++/CLI

```
property System.int PrimaryMemberType {  
    System.int get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

**Property Value**

Primary structure system member type as defined by [swStructureSystemMemberCreationType\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [PrimaryStructuralMemberFeatureData::PrimaryMemberType](#).

## Example

See the [ICornerManagementFolder](#) examples.

See the [IPrimaryMemberFacePlaneIntersectionFeatureData](#) examples.

See the [IPrimaryMemberPointLengthFeatureData](#) examples.

## See Also

[IPrimaryStructuralMemberFeatureData Interface](#)

[IPrimaryStructuralMemberFeatureData Members](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

# IPrint3DDialog (Type)

## IPrint3DDialog Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : IPrint3DDialog Interface

Allows access to the Print 3D dialog.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface IPrint3DDialog
```

Visual Basic (Usage)



```
Dim instance As IPrint3DDialog
```

C#

```
public interface IPrint3DDialog
```

C++/CLI

```
public interface class IPrint3DDialog
```

### Visual Basic for Applications (VBA) Syntax

See [Print3DDialog](#).

### Accessors

[IModelDocExtension::GetPrint3DDialog](#)

### Access Diagram

[Print3DDialog](#)

### See Also

[IPrint3DDialog Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)

## IPrint3DDialog Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IPrint3DDialog Interface

The following tables list the members exposed by [IPrint3DDialog](#).

### Public Methods

| Name                                                                                                             | Description                                                                 |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
|  <a href="#">UpdateDialog</a> | Updates the build statistics on the Print 3D dialog for a local 3D printer. |

[Top](#)

### See Also

[IPrint3DDialog Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

# IPrint3DDialog Interface Methods

[See Also](#)

|                                                                                   |                                                                                  |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|  | <a href="#">SolidWorks.Interop.sldworks Namespace</a> : IPrint3DDialog Interface |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|

For a list of all members of this type, see [IPrint3DDialog members](#).

## Public Methods

| Name                                                                                                           | Description                                                                 |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
|  <a href="#">UpdateDialog</a> | Updates the build statistics on the Print 3D dialog for a local 3D printer. |

[Top](#)

## See Also

[IPrint3DDialog Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

## UpdateDialog Method (IPrint3DDialog)

[See Also](#)

|                                                                                                                                         |                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|                                                        |  |
| <a href="#">SolidWorks.Interop.sldworks Namespace</a> > <a href="#">IPrint3DDialog Interface</a> : UpdateDialog Method (IPrint3DDialog) |                                                                                   |

Updates the build statistics on the Print 3D dialog for a local 3D printer.

## C# Syntax

Visual Basic (Declaration)

```
Function UpdateDialog() As System.Boolean
```

Visual Basic (Usage)  [Copy Code](#)

```
Dim instance As IPrint3DDialog  
Dim value As System.Boolean  
  
value = instance.UpdateDialog()
```

C#

```
System.bool UpdateDialog()
```

C++/CLI

```
System.bool UpdateDialog();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

True if the statistics are updated, false if not

## Visual Basic for Applications (VBA) Syntax

See [Print3DDialog::UpdateDialog](#).

## See Also

[IPrint3DDialog Interface](#)  
[IPrint3DDialog Members](#)

## Availability